Employee Performance Analysis — Project Report

1. Project Overview

Project Title: Employee Performance Analysis

This project analyzes employee data to understand workforce distribution, salary patterns, employee performance, locations, and employment status.

The project uses **Python, Excel, SQL, and Power BI** to perform data analysis and create an interactive dashboard.

---

2. Project Objective

The main objectives of this project are:

- Analyze the total number of employees.
- Understand employee distribution by department.
- Analyze average salary by department.
- Analyze employee performance scores.
- Identify employees with missing performance scores.
- Analyze employees by location.
- Analyze Active and Inactive employees.
- Create an interactive Power BI dashboard.
- Generate useful business insights from the data.

---

3. Technology Used

Technology	Purpose
Python	Data cleaning and analysis
Pandas	Data manipulation
NumPy	Data calculations
Matplotlib	Data visualization
Excel	Data analysis
SQL	Database queries and aggregation
Power BI	Interactive dashboard
DAX	KPI and calculated measures

---

4. Dataset Overview

The dataset contains **1,000 employee records**.

Important columns include:

- ID
- Name
- Age
- Gender
- Department
- Salary
- Joining Date
- Performance Score
- Experience
- Status
- Location
- Session

Department

The dataset contains three major departments:

- IT
- Sales
- HR

Employee Status

Employees are categorized as:

- Active
- Inactive

Location

Employees are distributed across:

- Los Angeles
- Chicago
- New York

---

5. Data Cleaning

The data was cleaned before performing the analysis.

The main data-quality issue identified was missing **Performance Score** values.

There are:

498 employees with missing Performance Scores.

These records should not be treated as valid performance scores when calculating the average performance.

Therefore, the dashboard displays the number of missing performance scores separately.

---

6. Key Performance Indicators

The Power BI dashboard contains four major KPIs:

Total Employees

1,000

Shows the total number of employees in the dataset.

Average Salary

₹5,920

Shows the average employee salary.

Average Performance Score

2.91

Shows the average recorded performance score.

Missing Performance Scores

498

Shows the number of employees for whom the performance score is missing.

---

7. Employee Analysis by Department

The employee distribution is relatively balanced across the three departments.

Department	Employees
IT	339
Sales	338
HR	323

Insight

IT has the highest number of employees, with **339 employees**, followed by Sales with **338** and HR with **323**.

The difference between departments is relatively small, indicating a balanced workforce structure.

---

8. Average Performance by Department

The average recorded performance score is:

Department	Average Performance
IT	3.73
Sales	2.76
HR	2.40

Insight

IT has the highest average performance score at 3.73.

Sales has an average score of **2.76**, while HR has the lowest average score at **2.40**.

However, because many employees have missing performance scores, these results should be interpreted carefully.

---

9. Average Salary by Department

The average salaries are approximately:

Department	Average Salary
HR	₹6,470
IT	₹6,170
Sales	₹5,190

Insight

HR has the highest average salary among the three departments.

Sales has the lowest average salary.

Salary differences can be investigated further using additional factors such as experience, job role, and employee level.

---

10. Employee Analysis by Location

Employees are distributed across three locations:

Location	Employees
Los Angeles	340
Chicago	333
New York	327

Insight

Los Angeles has the highest number of employees, followed by Chicago and New York.

The employee distribution across locations is also relatively balanced.

---

11. Employee Status Analysis

The dataset contains:

Status	Employees
Active	759
Inactive	241

Insight

The majority of employees are **Active**.

Approximately three-fourths of the workforce is active, while a smaller portion is inactive.

This metric can be useful for monitoring workforce retention and employee movement.

---

12. Performance Score Distribution

The dashboard contains a **Performance Score Distribution** chart.

Performance scores range from **1 to 5**.

The distribution helps identify how many employees fall into each performance category.

However, the dashboard separately identifies employees with missing scores so that missing values do not incorrectly appear as performance results.

---

13. SQL Analysis

SQL was used to perform various analytical operations, including:

- Counting total employees.
- Counting employees by department.
- Counting employees by location.
- Counting employees by status.
- Calculating average salary.
- Finding minimum salary.
- Finding maximum salary.
- Calculating average salary by department.
- Sorting departments based on salary.
- Retrieving selected employee records.

SQL Concepts Used

- `SELECT`
- `COUNT()`
- `AVG()`
- `MIN()`

- `MAX()`
- `GROUP BY`
- `ORDER BY`
- `LIMIT`

---

14. Power BI Dashboard

The Power BI dashboard provides an interactive overview of the employee dataset.

Dashboard Components

KPI Cards

- Total Employees
- Average Salary
- Average Performance Score
- Missing Performance Scores

Filters / Slicers

- Department
- Location
- Status

Charts

- Employees by Department
- Average Performance by Department
- Average Salary by Department
- Employees by Location
- Employee Status
- Performance Score Distribution

The slicers allow users to filter the dashboard and analyze specific departments, locations, or employee statuses.

---

15. Business Insights

Insight 1 — Workforce Distribution

The workforce is relatively evenly distributed among IT, Sales and HR.

Insight 2 — IT Performance

IT has the highest average recorded performance score.

Insight 3 — Salary

HR has the highest average salary, while Sales has the lowest average salary.

Insight 4 — Employee Status

There are significantly more Active employees than Inactive employees.

Insight 5 — Location

Los Angeles has the largest employee population among the three locations.

Insight 6 — Data Quality

The most important data-quality issue is the large number of missing performance scores.

498 employees do not have a Performance Score.

---

16. Recommendations

Based on the analysis, the following actions are recommended:

1. **Improve performance-score collection**
   Investigate why 498 employees have missing performance scores.
2. **Monitor department performance**
   IT's higher performance score can be studied to identify successful practices.
3. **Analyze salary differences**
   Salary should be analyzed together with experience, role and seniority.
4. **Monitor employee status**
   Track Active and Inactive employees over time to understand employee retention.
5. **Regularly update the Power BI dashboard**
   Refresh the dashboard when new employee data becomes available.
6. **Improve data quality**
   Establish standardized processes for collecting employee information.

---

17. Conclusion

The **Employee Performance Analysis** project demonstrates an end-to-end data analytics workflow.

The project includes:

Data → Cleaning → Analysis → SQL → Visualization → Power BI Dashboard → Business Insights

Python and Excel can be used for data preparation and exploratory analysis, SQL for querying and aggregation, and Power BI for creating an interactive dashboard.

The final dashboard provides management with a clear view of employee numbers, salary, performance, department distribution, locations and employee status.

The project also highlights an important data-quality issue: **498 employees have missing performance scores**, which should be addressed before making major performance-related decisions.